

1. Specifying timing constraints is a difficult but important step for designing embedded real-time systems. Give an example of each. Also, how would you derive (i) inter-task message deadlines from task deadlines, (ii) memory or disk I/O access deadlines from task deadlines, and (iii) task deadlines from end-to-end application deadlines. Construct a sporadic task along with an accompanying periodic task for an application laymen can understand.
2. Suppose a node accepts aperiodic real-time tasks from other nodes and soon encounters the arrival of its own real-time aperiodics whose deadlines can't be met locally by the node. In such a case, the late arriving aperiodics must be shipped out. This is obviously ineffective. Propose how to deal with this problem.
3. Write short notes on serial sensors and actuators
4. Describe in detail on sensor physics
5. In case of resource shortage because of failures, one may have to extend task periods or even skip tasks, e.g., execute the task once every two periods of the task. What does this imply in physical control systems and when and how to alter task periods during the operation of the controlled system?

6. Discuss on the digital signal processors (DSP)
7. Discuss on the Embedded Systems regarding on interrupts structure.
8. Describe in detail on architecture of embedded systems
9. Show that the priority ceiling protocol is not optimal. Do this by creating a task set which has unnecessary blocking, i.e., if we removed the blocking, the system would still operate correctly without any deadlock.
10. Other question
11. (Combined task assignment and scheduling) Is the system hazard higher or lower if the task precedence constraints are ignored and why? Explain how this information can be used to solve the combined task assignment and scheduling problem, and propose one more heuristic (other than relaxing the precedence constraints) to solve the problem.
12. Write an algorithm for generating the preferred list of receivers for a node in an n -dimensional hypercube. Transform the algorithm into a C program and submit the program execution results for $n = 4$ and $n = 6$.

13. Read and summarize anyone from the following:

1. One (1) paper on small proprietary real-time operating systems;
2. One (1) paper on real-time extensions of general-purpose OSs;
3. One (1) paper on low-power embedded OS/middleware; and
4. One (1) paper on low-power memory or disks

Again, your report must follow the format specified in the Assignments slides, and the main sources are annual IEEE RTSS and RTAS proceedings, ACM SOSP, USENIX OSDI and Annual Technical Conference proceedings. (You are NOT restricted to these sources!)